



Software Requirements Specification

Standard ISO/IEC/IEEE 29148 Compliant Blueprint

Project Name: BentaTrack: A Web-Based Inventory and Sales Management
System for Multi-Category Retail Management System

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Author / Organization: Van Paulino
Manuel Thomas Medina
Michael John Felices
John Moises Balingasa
Renz Rommel Dasal
Vanna Padilla
Pheche Prudenciado
Dawn Mabutin
Chezza Mae Gonzaga

Status: Draft / Approved





1. Introduction

1.1 Purpose

This system is being built for Estetika, a boutique store that currently tracks its products and sales using only a notebook and a calculator. This manual method makes it easy to lose track of items, forget updates, or misplace records.

The goal of this software is to give Estetika a simple, reliable way to manage inventory and sales without paper and manual counting. The strategic driver is convenience: a tool that is easy to learn, works with or without an internet connection, and reduces the daily effort of running the boutique by hand.

1.2 Intended Audience

Audience	Purpose
Instructor / Adviser	Reviews the requirements, scope, and completeness of the project.
Student Development Team	Uses this SRS to guide the design, development, testing, and documentation of the system. Team: Van Paulino, Manuel Thomas Medina, Michael John Felices, John Moises Balingasa, Renz Rommel Dasal, Vanna Padilla, Pheche Prudenciado, Dawn Mabutin, and Chezza Mae Gonzaga.
Owner / Administrator	Primary user; relies on the system for full control of products, sales, reports, and supplier information.
Inventory or Sales Staff	Uses the system daily for stock and sales tasks, within a limited, restricted access level.
Guest	Do not log in directly, but some features consider what they might want to see when browsing products.

1.3 Product Scope

The system focuses on two main areas, inventory management and sales management, supported by a simple dashboard and basic security features.

What the system will include

- Product records with name, code, category, brand, supplier, prices, stock, expiration date, image, and date added.
- Adding, editing, and deleting product records.
- Automatic stock deduction whenever a sale is recorded.
- Low stock pop-up notifications.
- An inventory change history log.





- Sales transaction recording, including quantity, price, date, staff, customer, discount, and payment method (Section 4.5).
- Daily, weekly, monthly, and yearly sales reports, plus best-selling products (Section 4.6).
- A dashboard showing totals, stock levels, and recent activity at a glance (Section 4.6).
- Product search by name or code, plus barcode scanning (Section 4.7).
- Role-based user accounts for the Owner/Administrator and Inventory/Sales Staff (Section 4.8).
- A simple, minimal interface in white and brown, with light and dark themes, the Estetika logo, and text labels with icons (Section 3.1).
- Support for desktop, laptop, tablet, and phone use, with offline operation and automatic sync (Section 4.9).

What the system will not include, at least for the initial version

- Automatic receipt generation.
- Credit or debit card payment processing.
- Delivery or rider assignment features, though raised as a possible future addition.
- QR code scanning; barcode-based lookup is the supported method (Section 4.7).

1.4 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification, the document describing what the system must do.
POS	Point of Sale, the part of the system that handles sales transactions.
GCash	A mobile wallet and online payment platform commonly used in the Philippines.
QR Code	A scannable code that could serve as an alternative to a barcode. Not implemented in this version.
Auto-sync	The process where offline transactions are automatically saved once the device regains internet access.
Estetika	The name of the boutique store this system is being developed for.



1.5 References

- Primary source: interview conducted by the student development team with a staff member of Estetika (“Ma’am Melanie”) during requirements gathering.
- No external design guidelines, APIs, or regulatory codes were used. Shopee and Lazada were mentioned only as casual examples of color and layout style, not as adopted standards.

2. Overall Description

2.1 Product Perspective

BentaTrack is a web application that is responsive across phones, tablets, laptops, and desktop computers. The Owner/Administrator uses it to manage products, monitor stock levels, and view sales reports, while Inventory/Sales Staff use it to record sales and update inventory throughout the day. A barcode scanner or device camera may be used to quickly look up a product during a transaction. The system also works without an internet connection, saving transactions locally on the device and automatically syncing once it reconnects.

System Flow: Staff Logs In -> Searches or Scans Product -> Records Sale -> System Deducts Stock Automatically -> Dashboard and Reports Update.

2.2 Product Functions

- Maintain accurate, always-available product and sales records.
- Warn the owner when stock is running low.
- Keep a history of inventory changes for reference.
- Automatically reduce stock the moment a sale is logged.
- Generate sales reports across different time periods.
- Show an at-a-glance dashboard summary whenever the system is opened.

2.3 User Classes and Characteristics

User	Main Tasks	Typical Device / Skill
Owner / Administrator	Monitors stock, sales, and overall store condition. Full access, including supplier information and account settings.	Desktop, laptop, tablet, or phone; basic to moderate computer skill.
Inventory or Sales Staff	Records sales and updates inventory day to day. Restricted from supplier contacts and password settings.	Desktop, laptop, tablet, or phone; basic computer or smartphone skill.

Guest (indirect)	Does not log in directly. Some features consider what they might want to see when browsing.	Not applicable.
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2.4 Operating Environment

Category	Requirement
System Type	Website, accessible through a standard web browser
Devices	Desktop computers, laptops, tablets, and cellphones
Internet	Works online and offline; offline data syncs automatically once reconnected
Client-Side Storage	IndexedDB (browser-native), for holding inventory/sales data locally while offline, before syncing to the central database
Optional Hardware	Barcode scanner or device camera for scanning product codes
Frontend	Next.js, Tailwind CSS, Chart.js, Dexie.js, next-pwa, html5-qrcode
Backend	Next.js, Node.js, NextAuth.js, bcrypt, Zod
Database	PostgreSQL, Prisma ORM
Deployment	Vercel

2.5 Design and Implementation Constraints

- Delivered as a functioning website, not a native mobile application.
- No automatic receipt generation in the initial version.
- Simple, minimal interface using white and brown as the primary color scheme.
- Must support both a light theme and a dark theme.
- Buttons and menus must use text labels together with icons, not icons alone.
- Must display the Estetika logo in an original, non-elaborate style.
- No specific programming language or data compliance standard required; left to the development team.

2.6 Assumptions and Dependencies

Assumptions

- Estetika will have at least periodic internet access for syncing.
- The owner will supply the official logo and branding materials.

- The store has, or will acquire, a compatible barcode scanner, or will use a smartphone camera instead.
- The Owner/Administrator and Inventory/Sales Staff have enough basic digital literacy to use a website-based system.

Dependencies

- Internet connection, for syncing offline data
- Web server and database
- Supported web browser
- File storage for product images
- Authentication system for login

3. External Interface Requirements

This section describes how the Estetika system will interact with users, hardware, other software, and networks.

3.1 User Interfaces

The interface shall be simple, minimal, and easy to learn, while still looking business appropriate for a boutique brand.

Screen / Page	Purpose
Login Page	Authenticates the user and routes them to the correct dashboard by role
Dashboard	Shows totals, stock levels, low stock alerts, recent transactions, and best sellers at a glance
Product / Inventory Management	Add, edit, delete, and view product records
Sales / Checkout	Record a sale, apply a discount, select a payment method, and complete a transaction
Sales Reports	View daily, weekly, monthly, and yearly sales, and best-selling products
Product Search	Look up a product by name, code, or barcode scan
Inventory History	View past inventory changes and purchase history for a product
User Accounts (Owner only)	Manage staff accounts and access levels



Design Requirements

- Light theme and dark theme options.
- White and brown as the primary and secondary colors.
- Estetika logo, in an original, simple design.
- Simple, minimal visual style.
- Text labels paired with icons on menus and buttons.
- Consistent layout across desktop, laptop, tablet, and phone.

3.2 Hardware Interfaces

The system is expected to run on common devices already available to the business, rather than specialized equipment.

- Barcode scanner, for identifying and retrieving product records.
- Standard desktop computers, laptops, tablets, and smartphones; no proprietary hardware required.

3.3 Software Interfaces

Interface	Purpose
Web Browser	Displays the Estetika system pages; no native mobile app required
Central Database	Stores product records, inventory data, sales transactions, and product images
Local Device Storage	Holds data temporarily while offline, before syncing to the central database
Barcode Scanner / Camera Input	Reads product barcodes for quick lookup
Notification Component	Shows pop-up alerts for low stock items



3.4 Communications Interfaces

- The system shall use a secure connection when exchanging data between a user's device and the central database.
- The system shall automatically synchronize offline transaction or inventory data with the central database once an internet connection becomes available.
- The system shall apply encryption when transmitting user credentials and business data over the network.
- The system shall inform the user once synchronization is complete, or if a connection issue prevents it from finishing.

4. System Features and Functional Requirements

4.1 Feature Framework Overview

Each feature below lists its priority level and the specific requirements the system must satisfy, based on the needs shared by the establishment staff during the interview.

4.2 Inventory Management

Priority: High

Lets the Owner/Administrator and Inventory/Sales Staff maintain accurate product records.

ID	Requirement	Priority
FR-001	The system shall allow authorized users to add a new product record.	High
FR-002	Each product record shall include product name, code, category, brand, supplier, purchase price, selling price, stock quantity, expiration date, image, and date added.	High
FR-003	The system shall allow editing of any existing product record.	High
FR-004	The system shall allow permanent deletion of a product record when a product is discontinued.	Medium
FR-005	The system shall automatically reduce the stored quantity of a product once a related sale is recorded.	High
FR-006	The system shall automatically label a product as "Out of Stock" when its quantity reaches zero, without deleting the record, so historical data is preserved (see Section 6.1).	High

4.3 Low Stock Alerts

Priority: High

Warns staff when a product's stock is running low, so resupply can be arranged before the item runs out.

ID	Requirement	Priority
FR-007	The system shall continuously monitor the stock quantity of every product.	High
FR-008	The system shall send a pop-up notification when a product reaches a low stock level.	High
FR-009	The system shall flag low stock products separately from products that are out of stock.	Medium

4.4 Inventory Change History

Priority: Medium

Keeps a record of inventory activity over time, so the owner can trace what happened to a product and when.

ID	Requirement	Priority
FR-010	The system shall log inventory changes with the date and time they occurred.	Medium
FR-011	The system shall record which product was affected by each change.	Medium
FR-012	The system shall keep a history of restocks, edits, removals, and customer purchases per product, including returns or refunds when applicable.	Medium

4.5 Sales Management

Priority: High

Records the details of every transaction at checkout, forming the basis for inventory deduction and sales reporting.

ID	Requirement	Priority
FR-013	The system shall record the product sold and the quantity sold for every transaction.	High
FR-014	The system shall record the unit price, total amount, date, and time of every transaction.	High
FR-015	The system shall record the staff member who processed the transaction.	High
FR-016	The system shall record customer information when provided.	Medium
FR-017	The system shall apply a discount to a transaction when one is given.	Medium
FR-018	The system shall record the payment method used, limited to cash, GCash, and bank transfer.	High
FR-019	The system shall not process credit card or debit card payments.	High
FR-020	Automatic receipt generation is not required for the initial version of the system.	Low

4.6 Sales Reporting and Dashboard

Priority: High

Gives the owner and staff quick visibility into sales performance and stock status.

ID	Requirement	Priority
FR-021	The system shall generate daily, weekly, monthly, and yearly sales reports.	High
FR-022	The system shall generate a best-selling products report.	High
FR-023	The system shall display total products, available stock, and low stock items on the dashboard for the Owner/Administrator.	High
FR-024	The system shall display recent transactions and best-selling products on the dashboard.	High
FR-025	The system shall present today's sales summary on request.	Medium

4.7 Product Search and Barcode Scanning

Priority: Medium

Helps staff locate a specific product quickly during daily operations.

ID	Requirement	Priority
FR-026	The system shall allow product search by product name.	Medium
FR-027	The system shall allow product search by product code.	Medium
FR-028	The system shall support barcode scanning to identify and retrieve a product record.	Medium
FR-029	QR code scanning is not required for this system.	Low

4.8 User Accounts and Security

Priority: High

Protects sensitive business data by giving different types of users different levels of access.

ID	Requirement	Priority
FR-030	The system shall require a password for every user account.	High
FR-031	The system shall grant the Owner/Administrator full access, including supplier contact information.	High
FR-032	The system shall grant Inventory/Sales Staff access to inventory and sales functions only, without access to supplier contacts or password settings.	High
FR-033	The system shall support two distinct levels of user access: Owner/Administrator and Inventory/Sales Staff.	High

4.9 Offline and Online Data Synchronization

Priority: Medium

Allows daily operations to continue even without an active internet connection.

ID	Requirement	Priority
FR-034	The system shall allow transactions to be recorded while offline.	Medium
FR-035	The system shall automatically sync offline data once an internet connection becomes available.	Medium
FR-036	The system shall not lose any transaction or inventory data recorded during offline use.	High

5. Non-Functional Requirements (NFRs)

This section describes how the system should behave, rather than what it should do, based on the needs shared by the store staff.

5.1 Performance Requirements

The store is small, with only a few users, so the system does not need to support heavy traffic. What matters most is speed and convenience during stock checks and sales.

- Dashboard and product pages should load in under 2 seconds under normal conditions.
- Product search, by name, code, or barcode, should return results almost instantly.
- Inventory numbers should update immediately after a sale is recorded.
- Performance should stay consistent whether the device is online or offline.

5.2 Security Requirements

Access to the system depends on the user's role, protecting sensitive business information while still letting staff do their work.

- Each user logs in with their own password.
- The Owner/Administrator has full access, including supplier contact details.
- Inventory/Sales Staff cannot view or change supplier information or account settings.
- Actions are tied to the account of the user who performed them.

5.3 Reliability and Availability

The system must keep working even without an internet connection, since the store cannot depend on always having stable wifi or data.

- The system continues accepting transactions and inventory updates while offline.
- Offline changes are saved on the device first and are not lost.
- Data automatically syncs with the main database once the connection is restored.

5.4 Software Quality Attributes

Usability is one of the most important qualities for this system, since ease of learning and use was specifically emphasized by the owner.

- Simple, minimal interface using text labels together with icons.
- White and brown color scheme, with both light and dark themes, and the Estetika logo.
- Consistent core features across desktop, laptop, tablet, and phone.
- Room to add future features, such as receipt printing or delivery tracking, without a full redesign.

6. Supplemental Artifacts

6.1 Database Schema Requirements

The database supporting the Estetika system centers on four main tables, summarized below, with role-based access rules tied to user accounts.

Table	Purpose
Product Table	Stores product details, pricing, stock levels, and images
Sales / Transaction Table	Records every sale, including items, pricing, discount, and payment method
Inventory History Table	Tracks restocks, returns, refunds, and other inventory changes over time
User Accounts Table	Stores login credentials and role-based access levels

Product Table

Field	Data Type	Description
Product Name	Text	Name of the product shown to customers and staff
Product Code	Text, unique	Identifier used for manual search and lookup
Barcode	Text or numeric	Value scanned for quick lookup
Category	Text	Grouping such as bags, accessories, or perfumes
Brand	Text	Manufacturer or brand of the product
Supplier	Text	Name and contact details of the supplying vendor; Owner/Administrator only
Purchase Price	Number, currency	Cost paid to the supplier per unit
Selling Price	Number, currency	Price shown to customers



Quantity in Stock	Number	Current available units; auto-decremented on each sale
Expiration Date	Date, optional	Applicable for products with a shelf life
Product Image	Image file	Photo used for identification and display
Date Added	Date	When the product was first recorded
Status	Text (Active / Low Stock / Out of Stock)	Updated automatically based on quantity in stock

Sales / Transaction Table

Field	Data Type	Description
Transaction ID	Unique ID	Identifier for each sale
Product Sold	Reference to Product Code	Product tied to the sale
Quantity Sold	Number	Units sold in the transaction
Unit Price	Number, currency	Price per unit at the time of sale
Discount	Number or percent, optional	Reduction applied, if any
Total Amount	Number, currency	Total price after discount
Date and Time	Date and time	When the transaction occurred
Staff Name	Text	Employee who processed the sale
Customer Information	Text, optional	Name or contact details, when provided
Payment Method	Text (Cash / GCash)	Credit and debit cards are not supported





Inventory History Table

Field	Data Type	Description
History ID	Unique ID	Identifier for each recorded change
Product Reference	Reference to Product Code	Product the change relates to
Change Type	Text (Restock / Edit / Return / Refund / Removal)	Type of inventory change recorded
Related Transaction	Reference to Transaction ID, optional	Linked sale, when applicable
Date and Time	Date and time	When the change occurred
Notes	Text	Short reason or remark about the change

User Accounts Table

Field	Data Type	Description
User ID	Unique ID	Identifier for each user account
Name	Text	Name of the account holder
Role	Enum	Owner/Administrator or Inventory/Sales Staff; determines access level
Password	Text, hashed	Login credential for the account

6.2 Appendices and Analytical Frameworks

The diagrams below illustrate how the sales process, inventory updates, user access levels, and offline syncing work step by step, based on the requirements in Sections 4, 5, and 6.1.



6.2.1 Sales Transaction Flow

Traces a single sale from product search through payment to the automatic deduction of stock.

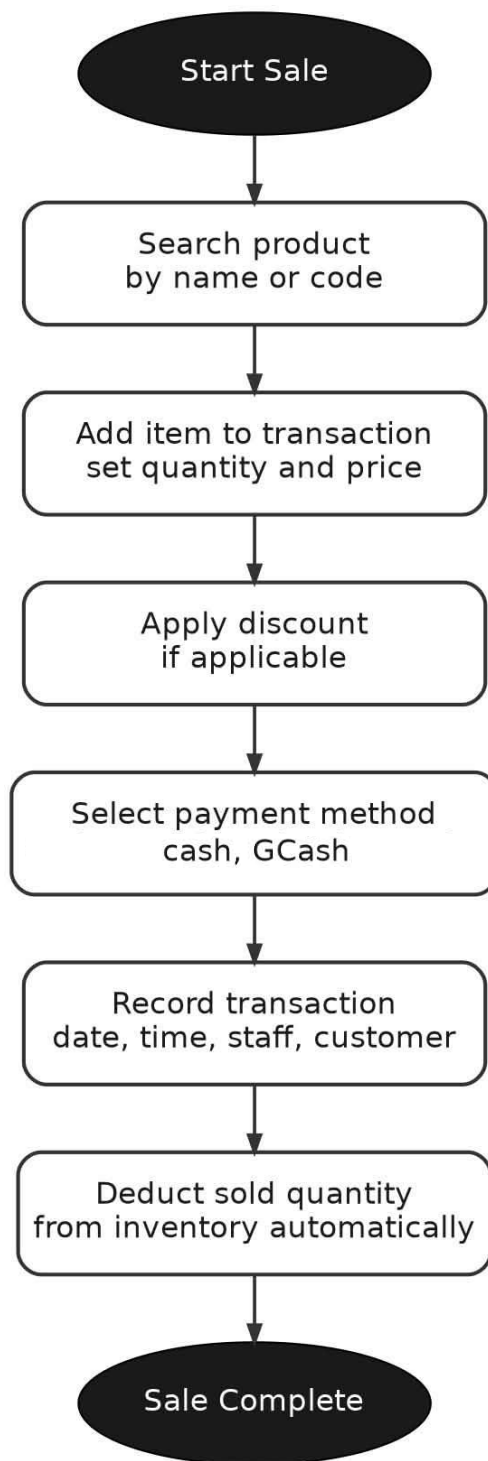


Figure 1. Sales Transaction Flow

6.2.2 Inventory Update Flow

Shows how staff add, edit, or remove a product, and how the system checks stock levels and sends a low stock alert.

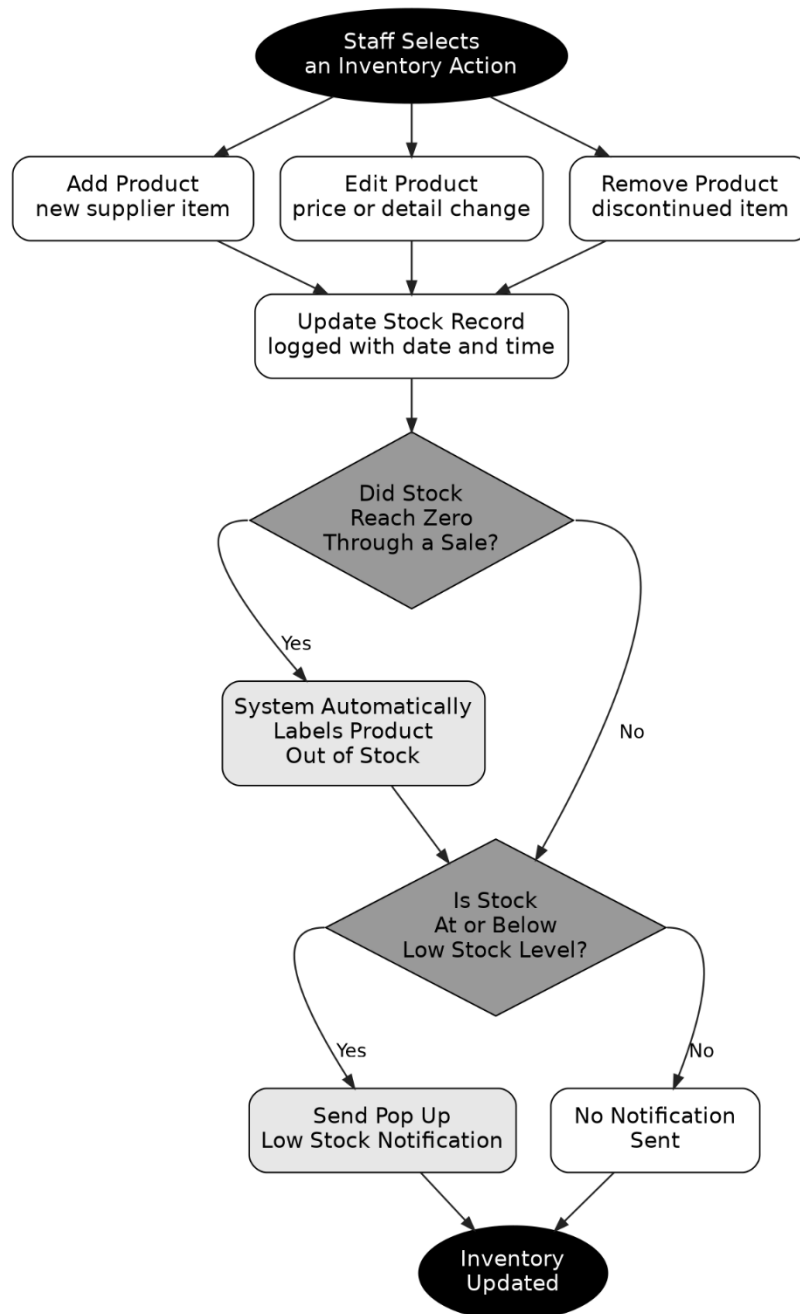


Figure 2. Inventory Update Flow

6.2.3 User Access Chart

Outlines what each role can view or edit, consistent with Sections 4.8 and 5.2.

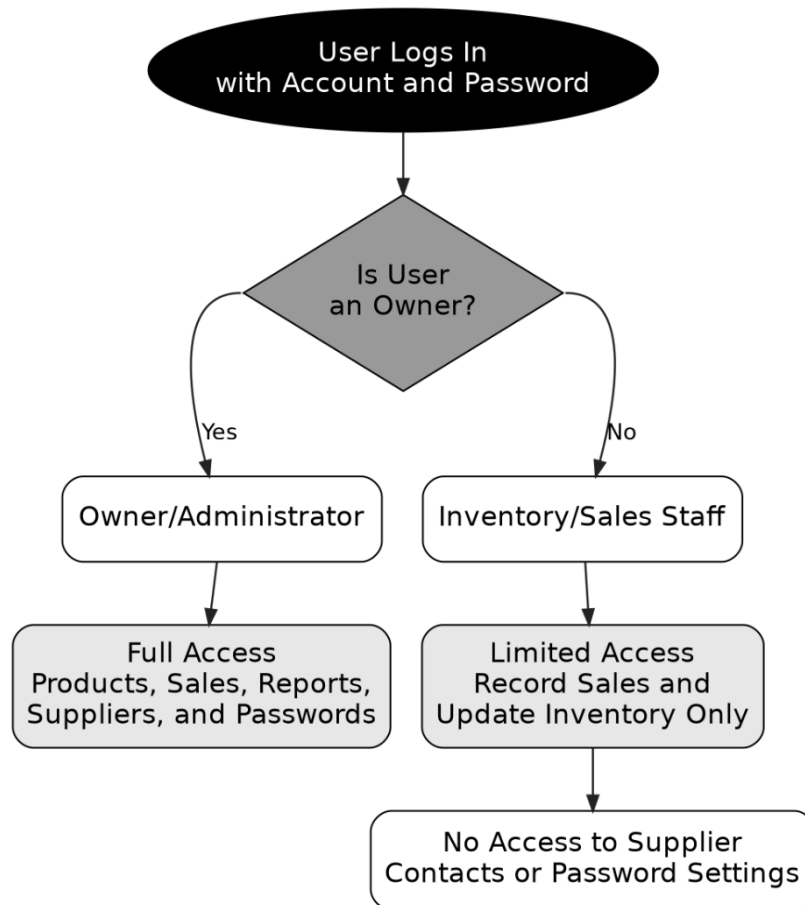


Figure 3. User Access Chart

6.2.4 Offline and Online Data Synchronization

Shows how a recorded sale or inventory update is handled depending on whether the device is online or offline.

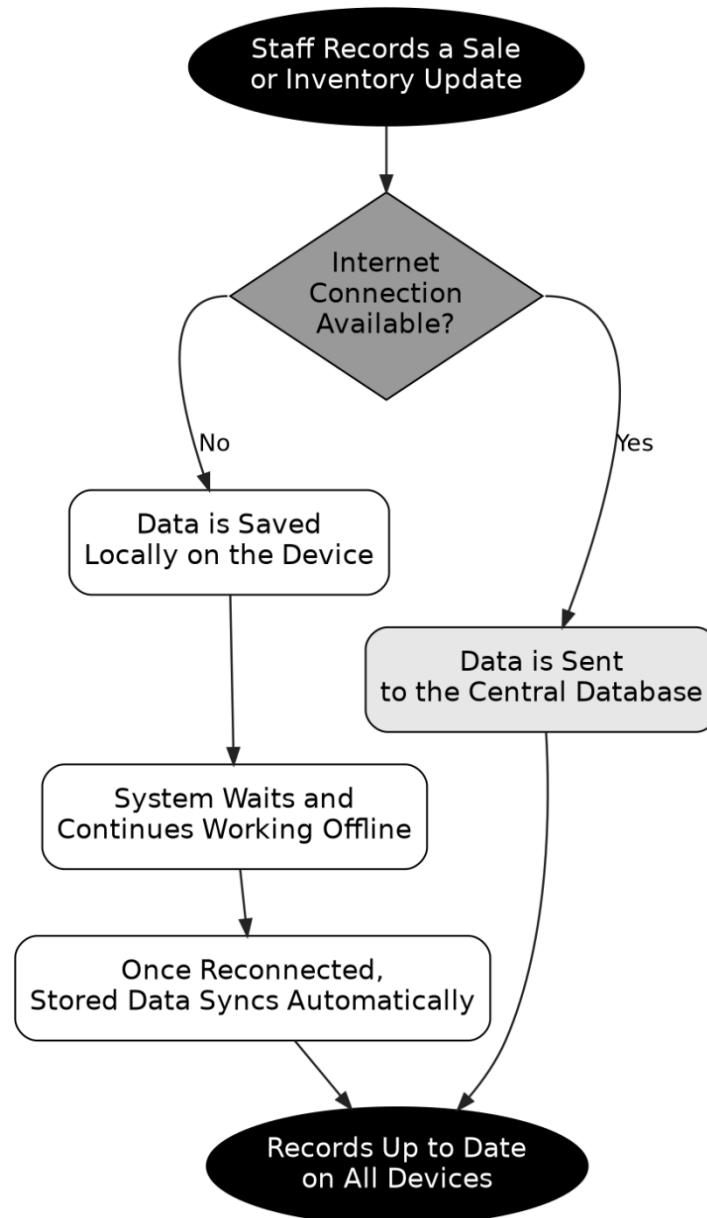


Figure 4. Offline-to-Online Data Synchronization



7. Approval and Sign-Off Matrix

The signatures below validate this Software Requirements Specification document as the official source of truth and development blueprint for the project build.

Role / Stakeholder	Name & Title	Signature / Date
Product Owner	Melanie Jabonite	
Team Leader	Manuel Thomas Medina	
UI Designer	Van Paulino	
Frontend Developer(s)	Dawn Mabutin	
	Chezza Mae Gonzaga	
Backend Developer(s)	Michael John Felices	
	John Moises Balingasa	
	Renz Rommel Dasal	
Database Designer(s)	Pheche Prudenciado	
	Vanna Padilla	

